

For the use of only a Registered Medical Practitioner or a Hospital or a Laboratory

Gemcitabine for Injection USP

GEMITA (Lyophilized)

DESCRIPTION:

Gemita for Injection USP is a white to off-white lyophilized powder or cake. It is soluble in water, slightly soluble in methanol, and practically insoluble in ethanol and polar organic solvents. After reconstitution, it becomes a clear colourless solution.

COMPOSITION:

Gemcitabine for Injection USP 200 mg

Each vial contains:

Gemcitabine Hydrochloride USP	
Equivalent to Gemcitabine	200 mg
Mannitol USP	200 mg
Sodium acetate USP	12.5mg
Hydrochloric acid USNF	to adjust pH
Sodium hydroxide USNF	to adjust pH

Gemcitabine for Injection USP 1g

Each vial contains:

Gemcitabine Hydrochloride USP	
Equivalent to Gemcitabine	1 g
Mannitol USP	1 g
Sodium acetate USP	62.5mg
Hydrochloric acid USNF	to adjust pH
Sodium hydroxide USNF	to adjust pH

Gemcitabine for Injection USP 1.4 g

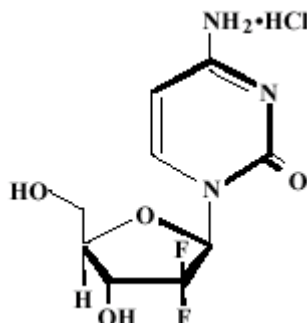
Each vial contains:

Gemcitabine Hydrochloride USP	
Equivalent to Gemcitabine	1.4 g
Mannitol USP	1.4 g
Sodium acetate USP	87.5mg
Hydrochloric acid USNF	to adjust pH
Sodium hydroxide USNF	to adjust pH

CHEMICAL STRUCTURE:

The Chemical name for Gemcitabine HCl is 2'-deoxy-2', 2'-difluorocytidine monohydrochloride (β - isomer). The empirical formula for Gemcitabine HCl is $C_9H_{11}F_2N_3O_4.HCl$. It has a molecular weight of 299.66.

The formulation is supplied in a sterile form for intravenous use only. Vials contain 200 mg, 1 g or 1.4 g of Gemcitabine HCl (expressed as free base) as a sterile lyophilized powder. The structural formula of Gemcitabine HCl is depicted below:



PHARMACOLOGY:

Mechanism of action

Gemcitabine kills cells undergoing DNA synthesis and blocks the progression of cells through the G1/S-phase boundary. Gemcitabine is metabolized by nucleoside kinases to diphosphate (dFdCDP) and triphosphate (dFdCTP) nucleosides. Gemcitabine diphosphate inhibits ribonucleotide reductase, an enzyme responsible for catalyzing the reactions that generate deoxynucleoside triphosphates for DNA synthesis, resulting in reductions in deoxynucleotide concentrations, including dCTP. Gemcitabine triphosphate competes with dCTP for incorporation into DNA. The reduction in the intracellular concentration of dCTP by the action of the diphosphate enhances the incorporation of gemcitabine triphosphate into DNA (self-potential). After the gemcitabine nucleotide is incorporated into DNA, only one additional nucleotide is added to the growing DNA strands, which eventually results in the initiation of apoptotic cell death.

Pharmacokinetics

Distribution

The volume of distribution was increased with infusion length. Volume of distribution of gemcitabine was 50 L/m² following infusions lasting <70 minutes. For long infusions, the volume of distribution rose to 370 L/m².

Gemcitabine pharmacokinetics are linear and are described by a 2-compartment model. Population pharmacokinetic analyses of combined single and multiple dose studies showed that the volume of distribution of gemcitabine was significantly influenced by duration of infusion and gender. Gemcitabine plasma protein binding is negligible.

Elimination

Metabolism

The active metabolite, gemcitabine triphosphate, can be extracted from peripheral blood mononuclear cells. The half-life of the terminal phase for gemcitabine triphosphate from mononuclear cells ranges from 1.7 to 19.4 hours.

Excretion

Gemcitabine disposition was studied in 5 patients who received a single 1000 mg/m²/30-minute infusion of radiolabeled drug. Within one (1) week, 92% to 98% of the dose was recovered, almost entirely in the urine. Gemcitabine (<10%) and the inactive uracil metabolite, 2'-deoxy- 2',2'-difluorouridine (dFdU), accounted for 99% of the excreted dose. The metabolite dFdU is also found in plasma.

Specific Populations

Geriatric Patients

Clearance of gemcitabine was affected by age and gender. The lower clearance in women and the elderly results in higher concentrations of gemcitabine for any given dose. Differences in either clearance or volume of distribution based on patient characteristics or the duration of infusion result in changes in half-life and plasma concentrations. Table 1 shows plasma clearance and half-life of gemcitabine following short infusions for typical patients by age and gender. Gemcitabine half-life for short infusions ranged from 42 to 94 minutes, and the value for long infusions varied from 245 to 638 minutes, depending on age and gender, reflecting a greatly increased volume of distribution with longer infusions.

Table 1: Gemcitabine Clearance and Half-Life for the “Typical” Patient

Age	Clearance Men (L/hr/m²)	Clearance Women (L/hr/m²)	Half-life⁺ Men (min)	Half-life⁺ Women (min)
29	92.2	69.4	42	49
45	75.7	57.0	48	57
65	55.1	41.5	61	73
79	40.7	30.7	79	94

⁺ Half-life for patients receiving a short infusion (<70 min)

Male and Female Patients

Females have lower clearance and longer half-lives than male patients as described in Table 1.

Patients with Renal Impairment

No clinical studies have been conducted with gemcitabine in patients with decreased renal function.

Patients with Hepatic Impairment

No clinical studies have been conducted with gemcitabine in patients with decreased hepatic function.

Drug Interactions

When gemcitabine (1250 mg/m² on Days 1 and 8) and cisplatin (75 mg/m² on Day 1) were administered in NSCLC patients, the clearance of gemcitabine on Day 1 was 128 L/hr/m² and on Day 8 was 107 L/hr/m². Data from patients with NSCLC demonstrate that gemcitabine and carboplatin given in combination does not alter the pharmacokinetics of gemcitabine or carboplatin compared to administration of either single agent; however, due to wide confidence intervals and small sample size, interpatient variability may be observed.

Data from metastatic breast cancer patients shows that gemcitabine has little or no effect on the pharmacokinetics (clearance and half-life) of paclitaxel and paclitaxel has little or no effect on the pharmacokinetics of gemcitabine.

INDICATIONS:

Gemcitabine is indicated for the treatment of locally advanced or metastatic non-small cell-lung cancer.

Gemcitabine is indicated for the treatment of adult patients with locally advanced or metastatic adenocarcinoma of the pancreas.

Gemcitabine is indicated for patients with 5-FU refractory pancreatic cancer.

Gemcitabine, in combination with paclitaxel, is indicated for the treatment of patients with unresectable, locally recurrent or metastatic breast cancer who have relapsed following adjuvant/neoadjuvant chemotherapy.

Prior chemotherapy should have included an anthracycline unless clinically contraindicated.

CONTRAINDICATIONS:

Gemcitabine is contraindicated in those patients with a known hypersensitivity to the gemcitabine. Reactions include anaphylaxis.

ADVERSE EFFECTS:

The most commonly reported adverse drug reactions associated with Gemcitabine treatment include: nausea with or without vomiting, raised liver transaminases (AST/ALT) and alkaline phosphatase, reported in approximately 60% of patients; proteinuria and haematuria reported in approximately 50% patients; dyspnoea reported in 10-40% of patients (highest incidence in lung cancer patients); allergic skin rashes occur in approximately 25% of patients and are associated with itching in 10% of patients.

The frequency and severity of the adverse reactions are affected by the dose, infusion rate and intervals between doses. Dose-limiting adverse reactions are reductions in thrombocyte, leucocyte and granulocyte counts.

Frequencies are defined as: Very common, Common), Uncommon, Rare, Very Rare

The following table of undesirable effects and frequencies is based on data from clinical trials. Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

System Organ Class	Frequency grouping
Blood and lymphatic system disorders	Very common • Leucopaenia (Neutropaenia Grade 3 = 19.3 %; Grade 4 = 6 %). Bone-marrow suppression is usually mild to moderate and mostly affects the granulocyte count • Thrombocytopaenia • Anaemia Common • Febrile neutropaenia Very rare • Thrombocytosis • Thrombotic microangiopathy
Infections and infestations	Common • Infections Not known • Sepsis
Immune system disorders	Very Rare • Anaphylactoid reaction
Metabolism and nutrition disorders	Common • Anorexia
Nervous system disorders	Common • Headache • Insomnia • Somnolence Very rare • Posterior reversible encephalopathy syndrome
Cardiac disorders	Rare • Myocardial infarct
Vascular disorders	Rare • Hypotension Very rare • Capillary leak syndrome
Respiratory, thoracic and mediastinal disorders	Very common • Dyspnoea –usually mild and passes rapidly without treatment Common • Cough • Rhinitis Uncommon • Interstitial pneumonitis • Bronchospasm –usually mild and transient but may require parenteral treatment
Gastrointestinal disorders	Very common • Vomiting • Nausea Common

	• Diarrhoea • Stomatitis and ulceration of the mouth • Constipation
Hepatobiliary disorders	Very common • Elevation of liver transaminases (AST and ALT) and alkaline phosphatase Common • Increased bilirubin Rare • Increased gamma-glutamyl transferase (GGT)
Skin and subcutaneous tissue disorders	Very common • Allergic skin rash frequently associated with pruritus • Alopecia Common • Itching • Sweating Rare • Ulceration • Vesicle and sore formation • Scaling Very rare • Severe skin reactions, including desquamation and bullous skin eruptions Not known • Pseudocellulitis
Musculoskeletal and connective tissue disorders	Common • Back pain • Myalgia
Renal and urinary disorders	Very Common • Haematuria • Mild proteinuria
General disorders and administration site conditions	Very common • Influenza-like symptoms - the most common symptoms are fever, headache, chills, myalgia, asthenia and anorexia. Cough, rhinitis, malaise, perspiration and sleeping difficulties have also been reported. • Oedema/peripheral oedema including facial oedema. Oedema is usually reversible after stopping treatment Common • Fever • Asthenia • Chills Rare • Injection site reactions-mainly mild in nature
Injury, poisoning, and procedural Complications	Radiation toxicity

Postmarketing experience (spontaneous reports) frequency not known (can't be estimated from the available data)

Nervous system disorders

Cerebrovascular accident, Posterior reversible encephalopathy syndrome (PRES)

Cardiac disorders

Arrhythmias, predominantly supraventricular in nature, Congestive heart failure, myocardial infarction

Blood and lymphatic system: Thrombotic microangiopathy

Vascular disorders

Peripheral vasculitis, gangrene, and capillary leak syndrome

Respiratory, thoracic and mediastinal disorders

Interstitial pneumonitis, pulmonary fibrosis, pulmonary edema, and adult respiratory distress syndrome, pulmonary eosinophilia

Gastrointestinal disorders

Ischaemic colitis

Hepatobiliary disorders

Serious hepatotoxicity, including liver failure and death, hepatic veno-occlusive disease

Skin and subcutaneous tissue disorders

Severe skin reactions, including desquamation and bullous skin eruptions, Lyell's Syndrome, Steven-Johnson Syndrome, Cellulitis, pseudocellulitis

Renal and urinary disorders

Renal failure

Haemolytic uraemic syndrome

Injury, poisoning and procedural complications

Radiation recall

DRUG INTERACTIONS:

No drug interaction studies have been conducted.

PRECAUTIONS AND WARNINGS:

Schedule-dependent Toxicity

Prolongation of the gemcitabine infusion time beyond 60 minutes or more frequent than weekly dosing resulted in an increased incidence of clinically significant hypotension, severe flu-like symptoms, myelosuppression, and asthenia. The half-life of gemcitabine is influenced by the length of the infusion.

Myelosuppression

Myelosuppression manifested by neutropenia, thrombocytopenia, and anemia occurs with gemcitabine as a single agent and the risks are increased when gemcitabine is combined with other cytotoxic drugs.

Pulmonary Toxicity and Respiratory Failure

Pulmonary toxicity, including interstitial pneumonitis, pulmonary fibrosis, pulmonary edema, and adult respiratory distress syndrome (ARDS), has been reported. In some cases,

these pulmonary events can lead to fatal respiratory failure despite discontinuation of therapy. The onset of pulmonary symptoms may occur up to 2 weeks after the last dose of gemcitabine. Permanently discontinue gemcitabine in patients who develop unexplained dyspnea, with or without bronchospasm, or evidence of severe pulmonary toxicity.

Hemolytic Uremic Syndrome

Hemolytic Uremic Syndrome to include fatalities from renal failure or the requirement for dialysis can occur in patients treated with gemcitabine. Most fatal cases of renal failure were due to HUS. Serious cases of thrombotic microangiopathy other than HUS have been reported with gemcitabine. Assess renal function prior to initiation of gemcitabine and periodically during treatment. Consider the diagnosis of HUS in patients who develops anemia with evidence of microangiopathic hemolysis, elevation of bilirubin or LDH, or reticulocytosis; severe thrombocytopenia; or evidence of renal failure (elevation of serum creatinine or BUN). Permanently discontinue gemcitabine in patients with HUS or severe renal impairment. Renal failure may not be reversible even with discontinuation of therapy. Renal failure may not be reversible even with discontinuation of therapy.

Hepatic Toxicity

Drug-induced liver injury, including liver failure and death, has been reported in patients receiving gemcitabine alone or in combination with other potentially hepatotoxic drugs. Administration of gemcitabine in patients with concurrent liver metastases or a pre-existing medical history or hepatitis, alcoholism, or liver cirrhosis can lead to exacerbation of the underlying hepatic insufficiency. Assess hepatic function prior to initiation of gemcitabine and periodically during treatment. Discontinue gemcitabine in patients that develop severe liver injury.

Embryo-fetal Toxicity

Based on animal data and its mechanism of action, gemcitabine can cause fetal harm when administered to a pregnant woman. Gemcitabine was teratogenic, embryotoxic, and fetotoxic in mice and rabbits. Advise pregnant women of the potential risk to a fetus. Advise females of reproductive potential to use effective contraception during treatment with gemcitabine and for 6 months after the final dose. Advise male patients with female partners of reproductive potential to use effective contraception during treatment with gemcitabine and for 3 months following the final dose.

Exacerbation of Radiation Therapy Toxicity

Gemcitabine is not indicated for use in combination with radiation therapy.

Concurrent (given together or ≤ 7 days apart)

Life-threatening mucositis, especially esophagitis and pneumonitis occurred in a trial in which gemcitabine was administered at a dose of 1000 mg/m² to patients with non-small cell lung cancer for up to 6 consecutive weeks concurrently with thoracic radiation.

Non-concurrent (given >7 days apart)

Excessive toxicity has not been observed when gemcitabine is administered more than 7 days before or after radiation. Radiation recall has been reported in patients who received gemcitabine after prior radiation.

Capillary Leak Syndrome

Capillary leak syndrome (CLS) with severe consequences has been reported in patients receiving gemcitabine as a single agent or in combination with other chemotherapeutic agents. Discontinue gemcitabine if CLS develops during therapy.

Posterior Reversible Encephalopathy Syndrome

Posterior reversible encephalopathy syndrome (PRES) has been reported in patients receiving gemcitabine as a single agent or in combination with other chemotherapeutic agents. PRES can present with headache, seizure, lethargy, hypertension, confusion, blindness, and other visual and neurologic disturbances. Confirm the diagnosis of PRES with magnetic resonance imaging (MRI) and discontinue gemcitabine if PRES develops during therapy.

Carcinogenicity, Mutagenicity and Impairment of Fertility

Long-term animal studies to evaluate the carcinogenic potential of gemcitabine have not been conducted. Gemcitabine was mutagenic in an in vitro mouse lymphoma (L5178Y) assay and was clastogenic in an in vivo mouse micronucleus assay.

Gemcitabine IP doses of 0.5 mg/kg/day (about 1/700 the human dose on a mg/m² basis) in male mice had an effect on fertility with moderate to severe hypospermatogenesis, decreased fertility, and decreased implantations. In female mice, fertility was not affected but maternal toxicities were observed at 1.5 mg/kg/day administered intravenously (about 1/200 the human dose on a mg/m² basis) and fetotoxicity or embryoletality was observed at 0.25 mg/kg/day administered intravenously (about 1/1300 the human dose on a mg/m² basis).

Gemcitabine is embryotoxic causing fetal malformations (cleft palate, incomplete ossification) at doses of 1.5 mg/kg/day in mice (approximately 0.005 times the recommended human dose on a mg/m² basis). Gemcitabine is fetotoxic causing fetal malformations (fused pulmonary artery, absence of gall bladder) at doses of 0.1 mg/kg/day in rabbits (about 0.002 times the recommended human dose on a mg/m² basis). Embryotoxicity was characterized by decreased fetal viability, reduced live litter sizes, and developmental delays.

Pregnancy

Gemcitabine can cause fetal harm when administered to a pregnant woman, based on its mechanism of action. Gemcitabine was teratogenic, embryotoxic, and fetotoxic in mice and rabbits. If this drug is used during pregnancy, or if a woman becomes pregnant while taking gemcitabine, the patient should be apprised of the potential hazard to a fetus.

Lactation

There is no information regarding the presence of Gemcitabine or its metabolites in human milk, or their effects on the breastfed infant or on milk production. Due to the potential for serious adverse reactions in breastfed infants from Gemcitabine, advise women not to breastfeed during treatment with Gemcitabine and for at least one week following the last dose.

Females and Males of Reproductive Potential

Pregnancy Testing

Verify pregnancy status in females of reproductive potential prior to initiating gemcitabine. Contraception gemcitabine can cause fetal harm when administered to a pregnant woman.

Females

Because of the potential for genotoxicity, advise females of reproductive potential to use effective contraception during treatment with gemcitabine and for 6 months after the final dose of gemcitabine.

Males

Because of the potential for genotoxicity, advise males with female partners of reproductive potential to use effective contraception during treatment with gemcitabine and for 3 months after the final dose.

Infertility

Males

Based on animal studies, gemcitabine may impair fertility in males of reproductive potential. It is not known whether these effects on fertility are reversible.

Pediatric Use

The safety and effectiveness of gemcitabine have not been established in pediatric patients. The maximum tolerated dose was 10 mg/m²/min for 360 minutes weekly for three weeks followed by a one-week rest period.

Geriatric Use

No overall differences in safety were observed between patients aged 65 and older and younger patients, with the exception of a higher rate of Grade 3-4 thrombocytopenia in older patients as compared to younger patients. Similar effectiveness was observed between older and younger women. There was significantly higher Grade 3/4 neutropenia in women 65 years of age or older. Gemcitabine clearance is affected by age, however there are no recommended dose adjustments based on patients' age.

Renal Impairment

No clinical studies have been conducted with gemcitabine in patients with decreased renal function.

Hepatic Impairment

No clinical studies have been conducted with gemcitabine in patients with decreased hepatic function.

Gender

Gemcitabine clearance is affected by gender. In single-agent studies of gemcitabine women, especially older women, were more likely not to proceed to a subsequent cycle and to experience Grade 3/4 neutropenia and thrombocytopenia.

DOSAGE AND ADMINISTRATION:

Adults:

i) Non small-cell lung cancer:

Single agent use: The recommended dose is 1000 mg/m^2 , given by intravenous infusion. The administration must be repeated once weekly for three weeks, followed by a one-week rest period. This four-week cycle is then repeated. A dose reduction or delay before each administration of the chemotherapy may be applied, based upon the amount of toxicity experienced by the patient.

Combination use: Gemcitabine in combination with cisplatin can be administered using two dosage regimens one regimen use a three-week schedule, the other uses a four-week schedule. The three-week schedule is the usual regimen; this three-week cycle uses gemcitabine 1250 mg/m^2 given by 30 minutes intravenous infusion on days 1 and 8, followed by one-week rest period. This three-week cycle is then repeated. A dosage reduction or delay before each administration of the chemotherapy may be applied, based upon the amount of toxicity experienced by the patient. The four-week cycle uses gemcitabine 1000 mg/m^2 given by 30 minutes intravenous infusion on days 1, 8 and 15, followed by one-week rest period. This four-week cycle is then repeated. A dosage reduction or delay before each administration of the chemotherapy may be applied, based upon the amount of toxicity experienced by the patient.

ii. Pancreatic adenocarcinoma:

The recommended dose is 1000 mg/m^2 , given by 30 minutes intravenous infusion This should be repeated once weekly for 7 weeks, followed by a week rest. Subsequent cycles should consist of injections once weekly for 3 consecutive weeks out of every 4 weeks. A dosage reduction or delay before each administration of the chemotherapy may be applied, based upon the amount of toxicity experienced by the patient.

iii) Breast cancer:

Paclitaxel (175 mg/m^2) administered on Day 1 over approximately 3 hours as an intravenous infusion, followed by gemcitabine (1250 mg/m^2) as a 30-minute intravenous infusion on Days 1 and 8 of each 21-day cycle. Dose reduction with each cycle or within a cycle may be applied based upon the amount of toxicity experienced by the patient.

For all indications:

Patients receiving gemcitabine should be monitored prior to each dose for platelet, leukocyte, and granulocyte counts and, if necessary, the dose of gemcitabine may be either reduced or withheld in the presence of hematologic toxicity, according to the following scale:

Absolute granulocyte count ($\times 10^6/\text{l}$) / Platelet count ($\times 10^6/\text{l}$) / % of total dose $>1,000$ and / $>100,000$ / 100 500 - 1,000 or / 50,000 - 100,000 / 75 < 500 or / $< 50,000$ / 0

Periodic physical examination and checks of renal and hepatic function should be made to detect non-hematologic toxicity. Dosage reduction with each cycle or within a cycle may be applied based upon the amount of toxicity experienced by the patient. Doses should be withheld until toxicity has resolved in the opinion of the physician.

Elderly patients:

Gemcitabine has been well tolerated by patients over 65 years of age. The pharmacokinetic data suggest that the metabolism of the drug is not affected by age.

Patients with hepatic or renal impairment:

Gemcitabine should be used with caution in patients with hepatic insufficiency or with impaired renal function as there is insufficient information from clinical studies to allow clear dose recommendation for this patient population. Mild to moderate renal insufficiency (GFR from 30mL/min to 80mL/min) has no consistent, significant effect on gemcitabine pharmacokinetics.

Children:

Gemcitabine has been studied in limited trials in children in a variety of tumor types. These studies did not provide sufficient data to establish the efficacy and safety of Gemcitabine in children.

Method of Administration:

Intravenous route.

Gemcitabine is well tolerated during infusion and is usually easy to administer. Reactions at the site of injection are rare: no case of cutaneous necrosis has been reported. If extravasation, the administration must be stopped immediately.

Preparation for Intravenous Infusion Administration**Reconstitution:**

- Reconstitute Gemcitabine for Injection USP 200 mg with 5ml of Sterile 0.9% w/v Sodium Chloride Injection, Gemcitabine for Injection USP 1g with 25ml of Sterile 0.9% w/v Sodium Chloride Injection and Gemcitabine for Injection USP 1.4g with 35 ml of sterile 0.9%w/v Sodium Chloride Injection. Shake gently to make a clear solution. These dilutions each yield a gemcitabine concentration of 38 mg/ml, which includes accounting for the displacement volume of the lyophilized powder (0.26 ml for 200 mg vial or 1.3 ml for 1 g vial or 1.82 ml for 1.4 g vial). The total volume upon reconstitution will be 5.26 ml or 26.3 ml or 36.82 ml, respectively. Complete withdrawal of vial contents will provide 200 mg or 1 g or 1.4 g of gemcitabine, respectively. It may further be diluted with 0.9% Sodium Chloride injection to concentration as low as 0.1 mg/ml. Due to solubility consideration, the maximum concentration for gemcitabine upon reconstitution is 40 mg/ml. Reconstitution at concentration greater than 40 mg/ml may result in incomplete dissolution and should be avoided. The reconstituted solutions should be used immediately or may be used within 24 hours if stored below 30°C.
- Discard unused portion.
- Solutions of reconstituted gemcitabine should not be refrigerated, as crystallization may occur.
- Inspect the reconstituted solution visually for particulate matter and discoloration prior to administration. If particulate matter or discoloration is found, do not administer.
- The compatibility of gemcitabine with other drug has not been studied. No

incompatibilities have been observed with infusion bottles or polyvinyl chloride bags and administration sets.

- Proper procedures for handling and disposal of cytotoxics should be followed for gemcitabine for injection USP.

OVERDOSE:

There is no known antidote for overdoses of gemcitabine. Myelosuppression, paresthesias, and severe rash were the principal toxicities seen when a single dose as high as 5700 mg/m² was administered by intravenous infusion over 30 minutes every 2 weeks to several patients in a dose-escalation study. In the event of suspected overdose, monitor with appropriate blood counts and provide supportive therapy, as necessary.

STORAGE:

Store below 30°C. Do not refrigerate.

INCOMPATIBILITIES:

No incompatibilities have been observed with infusion bottles or polyvinyl chloride bags and administration sets.

SHELF LIFE:

24 months

HANDLING AND DISPOSAL:

The normal safety precautions for cytostatic agents must be observed when preparing and disposing of the infusion solution. Pregnant personnel should not handle the product. Handling of the Injection should be done in a safety box and protective coats and gloves should be used. If no safety box is available, the equipment should be supplemented with a mask and protective glasses.

If the preparation comes into contact with the eyes, this may cause serious irritation. The eyes should be rinsed immediately and thoroughly with water. If there is lasting irritation, a doctor should be consulted. If the solution is spilled on the skin, rinse thoroughly with water.

Any unused product or waste material should be disposed of in accordance with local requirements.

PRESENTATION:

Gemcitabine for Injection USP 200 mg is available as sterile lyophilized powder for Injection in a 10 ml vial containing 200 mg of Gemcitabine.

Gemcitabine for Injection USP 1 g is available as sterile lyophilized powder for Injection in a 50 ml vial containing 1 g of Gemcitabine.

Gemcitabine for Injection USP 1.4 g is available as sterile lyophilized powder for Injection in 50 ml vial containing 1.4 g of Gemcitabine

REFERENCES:

1. Summary of Product Characteristics, Gemcitabine 1 g Powder for Solution for Infusion, Accord Healthcare Limited, 28/02/2019.
2. Prescribing Information, GEMZAR (gemcitabine for injection), Eli Lilly and Company, Indianapolis, IN 46285, USA. Revised on 05/2019.

MANUFACTURES INFORMATION:

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